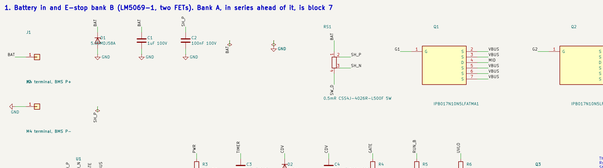
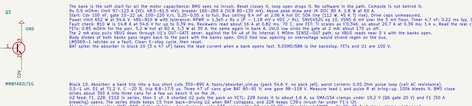
1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7







The bank is the soft start for all the motor capacitance BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it, so a compute crash can't drop the bank and kill the motors. Q1 and Q2 are the hot-swap FETs; Q3 is the CDV_B discharge FET.
Start: C4 100 nF, gate 10 nA, UVLO=20.5 V, OVLO=35.5 V, 0.5 mA to hold. Also setting 9.6 mV at 0.96 A for OC. 60A into 222 µF, 4.5 V sag, 2 msec. Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480–832 W with tolerance. RPWR = 1.5kΩ x R1 x (P - 1.1W mV + VO) / R2, SNSVLSZ=2 kΩ. VSNS 0 mV over the 3 mV floor. Timer 4.7 nF, 0.22 ms typ, 0.15–0.39 ms, 0.11–0.52 ms limits. Restart 1.1 sec typ, 0.8–1.5 sec limits. Retry 5 times, then latch off. Fault current 2.7 A, 2.4–3.0 A limits. 22.7 A at 0.59 ms. 1.4 A. Read the real curve at layout. FETs: 0.85 mOhm for the pair, 0.2 W hot at 60 A, 1.3 W at 80 A. A few amps again in bank A. UVLO low trips the gate at 2 mA. UVLO=33 is off.
The 2 mA also pulls VBUS down through U3's OUT–GATE zener, against the 54 uA of his internal 1 MOhm SENSE–OUT path, so VBUS reads near 0 V with the banks open.
Q3 discharges CDV_B, the 5 mF cap on the 54 V bus, when the banks open. UVLO tied low: opening on overvoltage would shield regen on the bus.
LM5069-1 latches on a fault. Clear: 3 retries in the time limit, then reset.
BAT spike: the absorber in block 10 (5 x 47 uF) takes the lead current when a bank opens fast. 5.05MΩ/58A is the backstop. FETs and U1 are 100 V.


Block 10, Absorber: a bank trip hits a bus short cuts 350–890 A. tools/absorber_sim.py. Jaack 54.6 V, no pack yet), worst corners: 0.05 Ohm pulse loop (cell AC resistance), 0.5–1 uH, 81 uF, 71.2 V, C –20 %, trip 8.8–17.5 ω. Three k7 uF cans give BAT 80–92 V. one gave 98–119 V. Measure lead L and pulse R at bring-up. 100k bleeds it. BMS close 12 msec after the bank. Bank 1 uF, 0.5 mΩ, 1 uH. Bus 1 uF, 4 mΩ, 0.5 uH.
U2 feed: F1, 20R, 1310 in series, then 1 uF. A shorted U2 pulls the pack on VCTL. 22R holds it to about 1.6 A, so SMAJ58A clamps under 19.2 V (20 with 20 V) and F1 (50 A breaking) opens. The series diode keeps CS from back-driving U2 when BAT collapses, and 220R keeps C39's inrush far under F1's 50 A breaking option (MI). DNP: if the 26 trips when it closes onto C38, remove the 5R and F1 (R = the delayed NMOS + 7k 8Ω 25 A at close, clamp unchanged (absorber_sim.py).
R19_A and R19_B diodes: a D1 pin 2–3 bridge or an F8 fault can't drive R19. F1 reaches UVLO through 847mW + 10K (2010) with a 12 V zener on FLT_A. F1' low holds UVLO near 0 V. Q4 and Q5 are 100 V, 0.5 A, 1.2 Ω, 1.5 pF. Gate 3.3 V, 4.7R, 2.2 nC. Turn-on 0.1 µs.
Converter: F2 20 A on the board end protects the pigtail (27 already limits at 9.7–12.5 A) tucks up a shorted Q7). The OVD has no reverse protection, so the keyed connector carries polarity. Q7 PGD high (good) turns on the 2N7002, which pulls ON/OFF to –Vin through 220R on (5.45 V at 2 mA), 220R limits a pin 5–11 short to 55 mA; 470k and the 12 V rail limit a pin 5–12 short to 25 µA. Q7 is 60 V, 115 mA, 7.5 Ω, 3 pF.
1 uF on BST_BAT SMBJ, a reset tied at power-up no longer arms the latch (tools/latch_sim.py, 36 cases). Reset interlock (MC): Q15 shorts RST_SRC until MID is over about 11 V (1.0 nmol, Q15 with Q14 shorts the gate to source). Q14 is 60 V, 0.3 A, 3 Ω, 1.5 pF, 1.5 V threshold, 1.5 nA at 50 V, 0.5 nA at 25 V. Q14 drain to RST_SRC 100k, source to gate 1M. Q15